

Software Engineering

Lab 3

Q. 36 Restaurant Table Booking & Pre-Ordering App

Name: Kallepalli Sai Meghana

Section: E

SRN: PES1UG24AM911

Architecture Selection:

Client-Server Architecture

Architecture	Advantages	Disadvantages	Suitability
Layered	Easy to understand, separation of concerns	Changes can affect multiple layers	
Microservices	Independent scaling, fault isolation	Complex to develop/deploy	
Client-Server	Centralized data, simple deployment, easy consistency	Server can become bottleneck	This is chosen for this lab.

I chose Client-Server Architecture for the Restaurant Table Booking & Pre-Ordering App.

1. Architectural Choice: The proposed system uses Client-Server Architecture. The Diner App and Restaurant Management Dashboard act as clients, while the server-side components manage reservations, pre-orders, menu and table information, payments, and restaurant data. A centralized Restaurant Database stores the system information.

2. Reason 1 - Centralized Reservation Management: The application needs accurate and up-to-date information about table availability and reservations. A centralized server allows all diners and the restaurant manager to access the same reservation data. This helps prevent conflicting bookings and keeps table availability consistent.

3. Reason 2 - Simple Communication and Management: The Diner App can communicate with the server to view menus, check table availability, make reservations, submit pre-orders, and make payments. At the same time, the Restaurant Management Dashboard can communicate with the server to manage reservations and restaurant operations. This makes the architecture simple and suitable for the application.

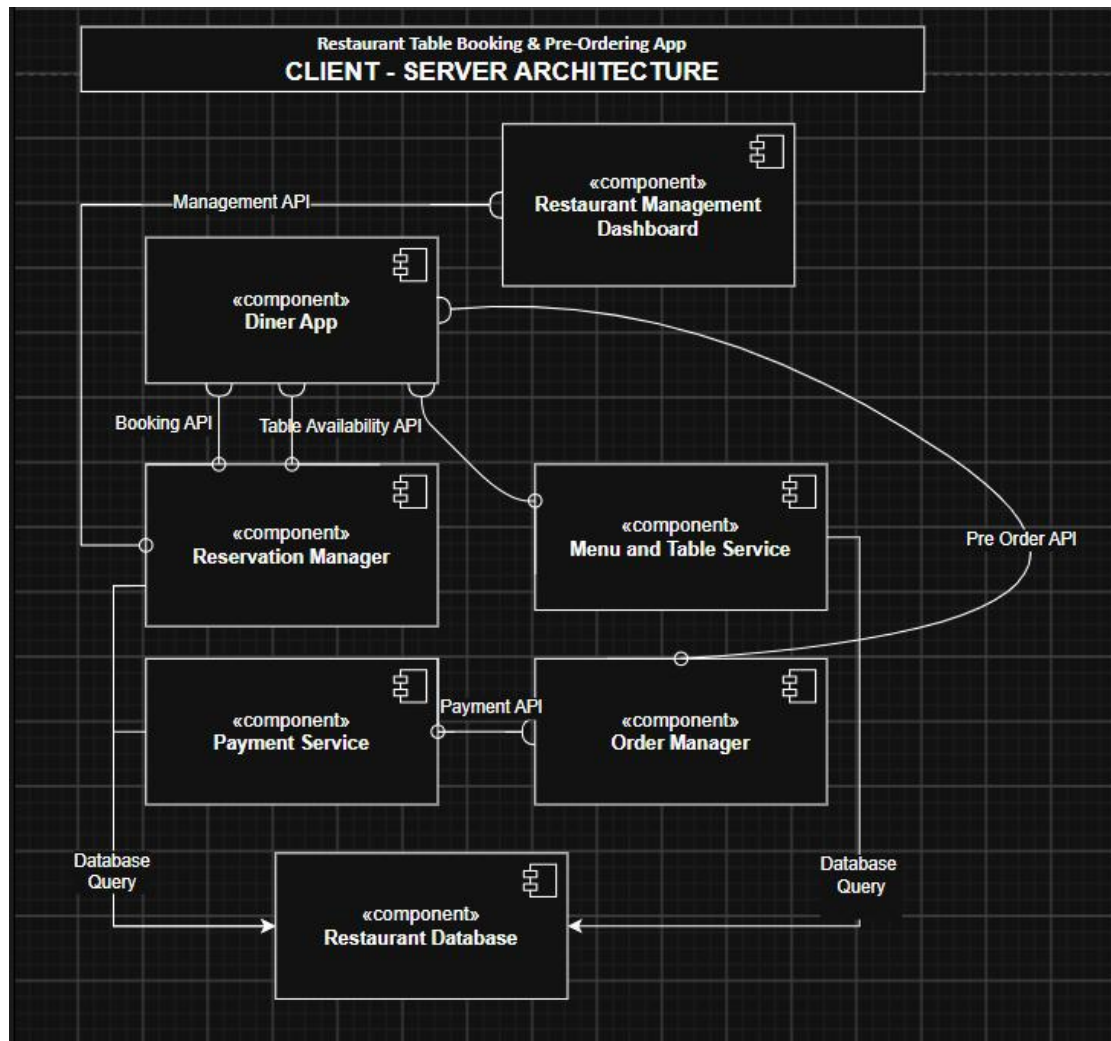
4. Security Advantage: Security can be managed centrally at the server. Authentication and authorization can control access based on the user's role. For example, diners can manage their own bookings and orders, while restaurant managers can access management functions. Payment processing can also be handled separately by the Payment Service.

5. Performance Benefit: The centralized server can efficiently manage and retrieve reservation, menu, order, and table information from the database. Frequently requested

information such as menu data can also be cached, reducing repeated database access and improving response time.

Conclusion: Therefore, Client-Server Architecture is appropriate because it provides centralized data management, simple communication between users and restaurant services, controlled security, and efficient access to restaurant information.

Component Diagram:



Components - The system contains 7 components:

1. Diner App

- View menu
- Check table availability
- Book a table
- Pre-order food
- Make payment

2. Restaurant Management Dashboard

- Manage reservations
- Monitor restaurant operations
- Manage tables and orders

3. Reservation Manager

- Table availability
- Creating reservations
- Updating reservations
- Cancelling reservations

4. Menu & Table Service

- Menu information
- Food prices
- Table information and availability

5. Order Manager

- Pre-orders
- Order details
- Order totals
- Communication with the payment service

6. Payment Service

- Payment processing
- Payment status
- Transaction information

7. Restaurant Database

- Diner information
- Reservation information
- Table information
- Menu information
- Order information
- Payment records